Josef Bisits

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Teaching experience

Mathematics

- **Tutoring** — UNSW, Sydney
  I have been involved in mathematics tutoring and marking since my honours year at UNSW including: — 2021–Current
  - Mathematics Drop in Centre;
  - Mathematics 1A (MATH1131/1141);
  - Mathematics 1B (MATH1231/1241);
  - Numerical Methods and Statistics (MATH2089);
  - Introduction to Atmosphere and Ocean Dynamics (MATH2241);
  - Fluids, Oceans, and Climate (MATH3261-5285), developed computer lab content; and
  - Third year project (MATH3001) marker.
- **Student supervision**
  - Krishiv Kukreja, research intern (co supervisor with A/Prof. Jan Zika) — July-September 2026.

Music

- **Private and ensemble double bass tuition** — Sydney
  During my double bass career I tutored private students and youth orchestra ensembles including: — 2013–2022
  - Sydney Youth Orchestra ensembles;
  - Penrith Youth Orchestra; and
  - Playerlink, run by the Sydney Symphony Orchestra.

Scientific Experience

Research positions

- **Postdoctoral research associate**, Australian Centre for Excellence in Antarctic Science — UNSW, Sydney
  Diagnosing numerical mixing in finite volume models of fluid flow with A/Prof. Jan D. Zika. — 2025-current
- **Research assistant**, School of Mathematics and Statistics — UNSW, Sydney
  Numerical modeling and analysis of tracer release experiments in turbulent flows with A/Prof. Jan Zika. — 2022
- **Research assistant**, Faculty of Built Environment — UNSW, Sydney
  Historical water quality data analysis to develop a predictive water quality model with A/Lec. Simon Lloyd. — 2022

Industry positions

- **Ocean modelling contractor** — Submarine
  Build and validate a model to simulate tracer release experiments in Port Angeles Harbour, Washington, USA. — 2025-2026

Education

- **Doctor of Philosophy**, The University of New South Wales — Sydney
  School of Mathematics and Statistics — 2022–2025
  - Supervisors: Associate Professor Jan Zika, Professor Trevor McDougall and Dr Taimoor Sohail
  - Thesis: Non-linear controls on ocean circulation and mixing in the high-latitude oceans
  - Published version of PhD thesis abstract
- **Bachelor of Science (Advanced mathematics, honours)**, The University of New South Wales — Sydney
  School of Mathematics and Statistics — 2016–2021
  - Major: Applied mathematics
  - Supervisors: Associate Professor Jan Zika and Dr Geoff Stanley
  - Thesis: Passive tracer mixing along isopycnal surfaces in turbulent flows
  - Honours grade: class 1
- **Bachelor of Music**, The University of Sydney — Sydney
  Sydney Conservatorium of Music — 2007–2010
  - Major: Performance, double bass
  - Double bass instructor: Alex Henery

Publications

- **Bisits, Josef I.** et al. "A simple formula to infer rates of numerical mixing in finite volume ocean models part two: application". In prep.
- Zika, Jan D., **Bisits, Josef I.** et al. "A simple formula to infer rates of numerical mixing in finite volume ocean models part one: theory and example". In prep.
- **Bisits, Josef I.**, McDougall, Trevor J., and Zika, Jan D. Aug. 2026. "Determining how nonlinearities in the equation of state affect diffusive interfaces in the ocean". In: *Journal of Fluid Mechanics* 1040, A33. DOI: 10.1017/jfm.2026.11889.
- Lloyd, Simon D., **Bisits, Josef I.** et al. 2026c. "Swimming in urban estuaries: understanding stormwater contamination events and recovery from historical data". In: *Water Research*, p. 125686. DOI: 10.1016/j.watres.2026.125686.
- **Bisits, Josef I.**, Zika, Jan D., and Sohail, Taimoor. May 2025. "Cabbeling as a catalyst and driver of turbulent mixing". In: *Journal of Fluid Mechanics* 1011, A17. DOI: 10.1017/jfm.2025.349.
- **Bisits, Josef I.**, Zika, Jan D., and Evans, Dafydd Gwyn. Dec. 2024. "Does Cabbeling Shape the Thermohaline Structure of High-Latitude Oceans?" In: *Journal of Physical Oceanography* 54 (12), pp. 2419–2430. DOI: 10.1175/JPO-D-24-0061.1.
- **Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. Feb. 2023. "Can We Accurately Quantify a Lateral Diffusivity from a Single Tracer Release?" In: *Journal of Physical Oceanography* 53 (2), pp. 647–659. DOI: 10.1175/JPO-D-22-0145.1.

Selected conference proceedings and seminars

Oral presentations

- **Bisits, Josef I.**, Zika, Jan D., and Sohail, Taimoor. Feb. 2026. "Cabbeling as a catalyst and driver of turbulent mixing". In: *Ocean Sciences*.
- **Bisits, Josef I.** Oct. 2025. "Non-linear controls on ocean circulation and mixing in the high-latitude oceans". In: *Australis Centre for Excellence in Antarctic Science seminar series*.
- **Bisits, Josef I.**, Zika, Jan D., and Evans, Daffyd Gwynn. June 2024. "Does cabbeling shape the thermohaline structure of high-latitude oceans?" In: *Gordon Research Seminar on Ocean Mixing*.

- **Bisits, Josef I.**, Zika, Jan D., and Evans, Daffyd Gwynn. Feb. 2024. “Does cabbeling shape the thermohaline structure of high-latitude oceans?” In: *Australian Meteorological and Oceanographic Society annual conference*.
- **Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. July 2023. “Can we accurately quantify a lateral diffusivity from a single tracer release?” In: *XVIII General Assembly of the International Union of Geodesy and Geophysics (IUGG*.
- **Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. Nov. 2022. “Can we accurately quantify a lateral diffusivity from a single tracer release?” In: *Australian Meteorological and Oceanographic Society annual conference*.

Poster presentations

- **Bisits, Josef I.**, Zika, Jan D., and Sohail. June 2024. “The effect of non-linear processes on mixing”. In: *Gordon Research Conference on Ocean Mixing*.
- **Bisits, Josef I.**, Zika, Jan D., and Sohail, Taimoor. Nov. 2024. “The effects of cabbeling on mixing and energetics in polar oceans”. In: *Australian Antarctic Research Conference*.
- **Bisits, Josef I.**, Zika, Jan D., and Evans, Daffyd Gwynn. July 2023. “Does cabbeling shape the thermohaline structure of high-latitude oceans?” In: *Physics of the Ocean Summer School*.
- **Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. Nov. 2022. “Can we accurately quantify a lateral diffusivity from a single tracer release?” In: *Australian Centre for Excellence in Antarctic Science workshop*.

Programming Skills

I have wide experience in programming which includes:

- using High Performance Computing systems to run models on both CPU’s and GPU’s;
- building models for ocean and fluid dynamics simulations; and
- data visualisation.

My programming language of choice is julia. Some packages I am involved with are:

- Author `StaircaseShenanigans.jl`, `TwoLayerDirectNumericalShenanigans.jl`, `RasterHistograms.jl`;
- Developer `PassiveTracerFlows.jl`; and
- Contributor `FourierFlows.jl`, `GeophysicalFlows.jl`, `Oceananigans.jl` amongst others.

I am also fluent with , , , , LaTeX and git.

Selected outreach activites

- **Helper for Do the Maths!**, *University of New South Wales.* 2022, 2023
- **Helper for Experience UNSW Science day**, *University of New South Wales.* 2022
- **Designed the Climate@maths display for UNSW open day**, *University of New South Wales.* 2022, 2023, 2024
- **Helper for uDASH work experience workshop**, *University of New South Wales.* 2022

Professional activities

- **Peer reviewer**
  Carried out peer reviews of articles submitted to: 2023-current
  - Journal of Geophysical Research: Oceans;
  - EGUSphere;
  - Ocean Modelling.
- **Conference session chair** UNSW, Sydney
  School of Mathematics and Statistics postgraduate conference. 2024

Scholarships and Awards

- Best applied mathematics talk at UNSW School of Mathematics and Statistics postgraduate conference. 2024
- Michael Tallis Research travel award. 2023
- Australian Government Research Training Program Scholarship. 2022–2025
- School of Mathematics and Statistics research scholarship top-up. 2022–2025
- Australian Centre for Excellence in Antarctic Research PhD student. 2022–2025
- Michael Bannigan scholarship for academic achievement. 2010
- Corina d'Hage string scholarship. 2008–2010
- Winner of Sydney Youth Orchestra concerto competition. 2006

Orchestral Experience

Professional

- **Sydney Symphony Orchestra** Sydney
  Regular casual double bassist, full time contract musician in 2014, 2015 and September 2017–September 2018. 2009–current
- **Australian Chamber Orchestra** Sydney
  Regular casual double bassist. 2013–current
- **Opera Australia Orchestra** Sydney
  Regular casual double bassist. 2016–current
- **Click Track Music** Sydney
  Regular casual double bassist. 2009–current
- **Tasmanian Symphony Orchestra** Tasmania
  Casual double bassist. 2013–2018
- **Auckland Philharmonia** Auckland
  Casual double bassist. 2013
- **Ulster Orchestra** Belfast
  Held a trial for double bass section leader. Winter 2012

Training programs

- **Emerging Artist**, Australian Chamber Orchestra Sydney
  Mentor: Maxime Bibeau. 2013
- **Southbank Sinfonia** London
  Principal double bassist, twice appeared as a soloist. 2011-2012
- **Sydney Symphony Fellowship**, Sydney Symphony Orchestra Sydney
  Also appeared with Sydney Sinfonia from 2008-2010. 2010
- **Australian Youth Orchestra** Australia
  Double bass section player. 2007–2008
- **Sydney Youth Orchestra** Sydney
  Principal double bassist, appeared as double bass soloist in 2006. 2006–2008

References

- **Associate Professor Jan Zika,** *University of New South Wales, Sydney.* Email: j.zika@unsw.edu.au
- **Dr Taimoor Sohail,** *Research Fellow, University of Melbourne.* Email: taimoor.sohail@unimelb.edu.au
- **Emeritus Professor Trevor McDougall,** *University of New South Wales, Sydney.* Email: trevor.mcdougall@unsw.edu.au
- **Dr Geoff Stanley,** *Canadian Centre for Climate Modelling and Analysis.* Email: geoff.stanley@ec.gc.ca